

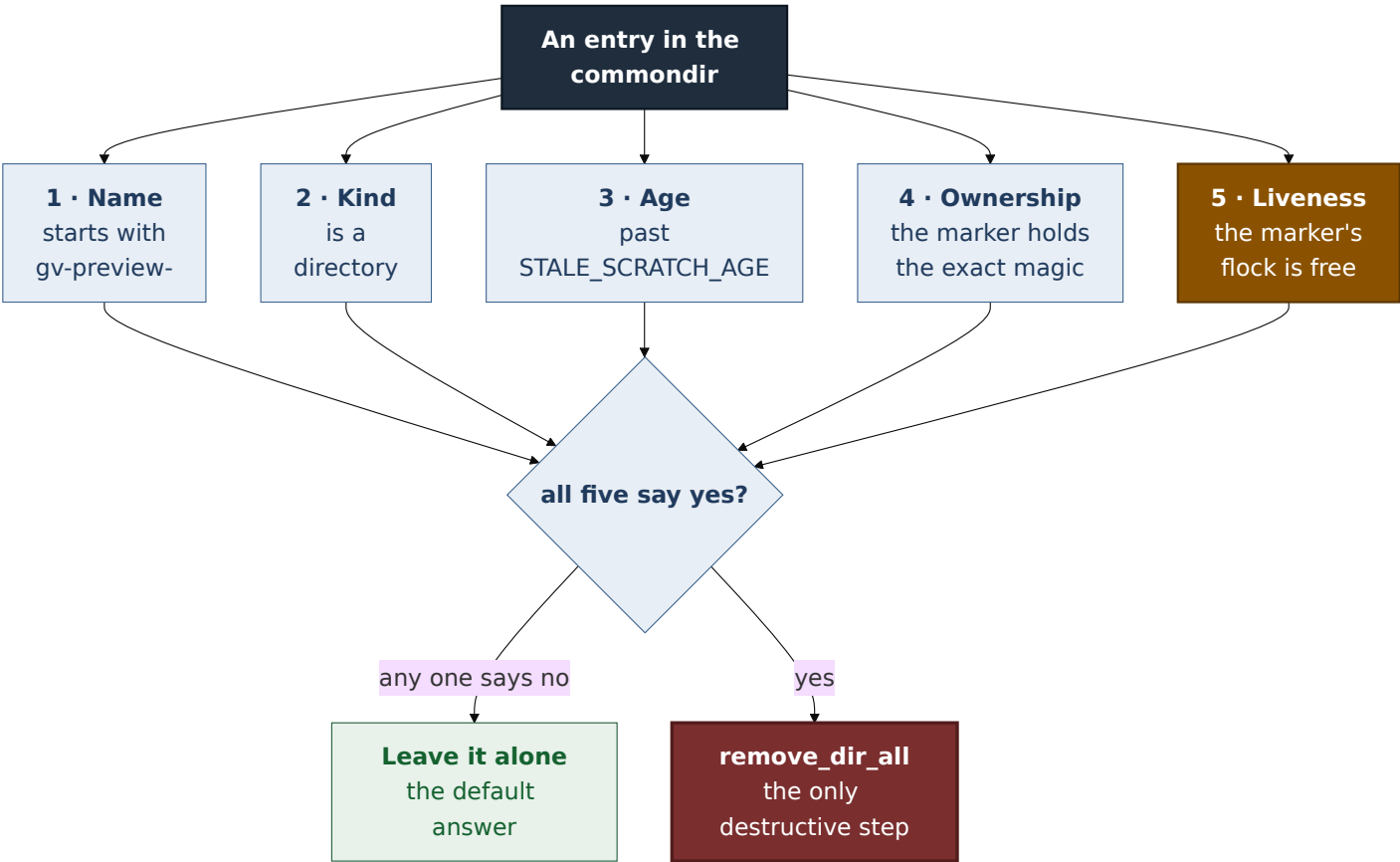
ADR 0111 — One refusal is not evidence of an owner

- **Status:** Accepted — implemented, mutation-proved two ways per test
- **Date:** 2026-09-03
- **Issue:** #598
- **Extends:** [ADR 0104](#)
- **Supersedes / superseded by:** —

Context

A preview builds its merge in a **scratch store**: a `gv-preview-*` directory inside the served repository's `.git`. When a preview finishes, its store goes with it. When a preview is killed, the store stays, and `sweep_stale` is what reclaims it on the next preview.

Deleting directories inside a user's `.git` is the most destructive thing this module does, so the sweep was built to refuse by default. A directory is removed only when every gate answers yes:



They are evaluated in that order, cheapest first, and the first `no` ends the question — so gate 5 is reached only by a directory that already looks abandoned on every count a filesystem can answer.

Gate 5 is the interesting one. Age is a timestamp, not a lease: a preview whose store was created two hours ago is indistinguishable from a corpse by age alone. The advisory `flock` on the marker is what tells them apart, because the kernel releases it exactly when the owning process goes away — which is the question the sweep is really asking.

Gate 5 was one call:

```
f.try_lock().ok()?;
```

What went wrong

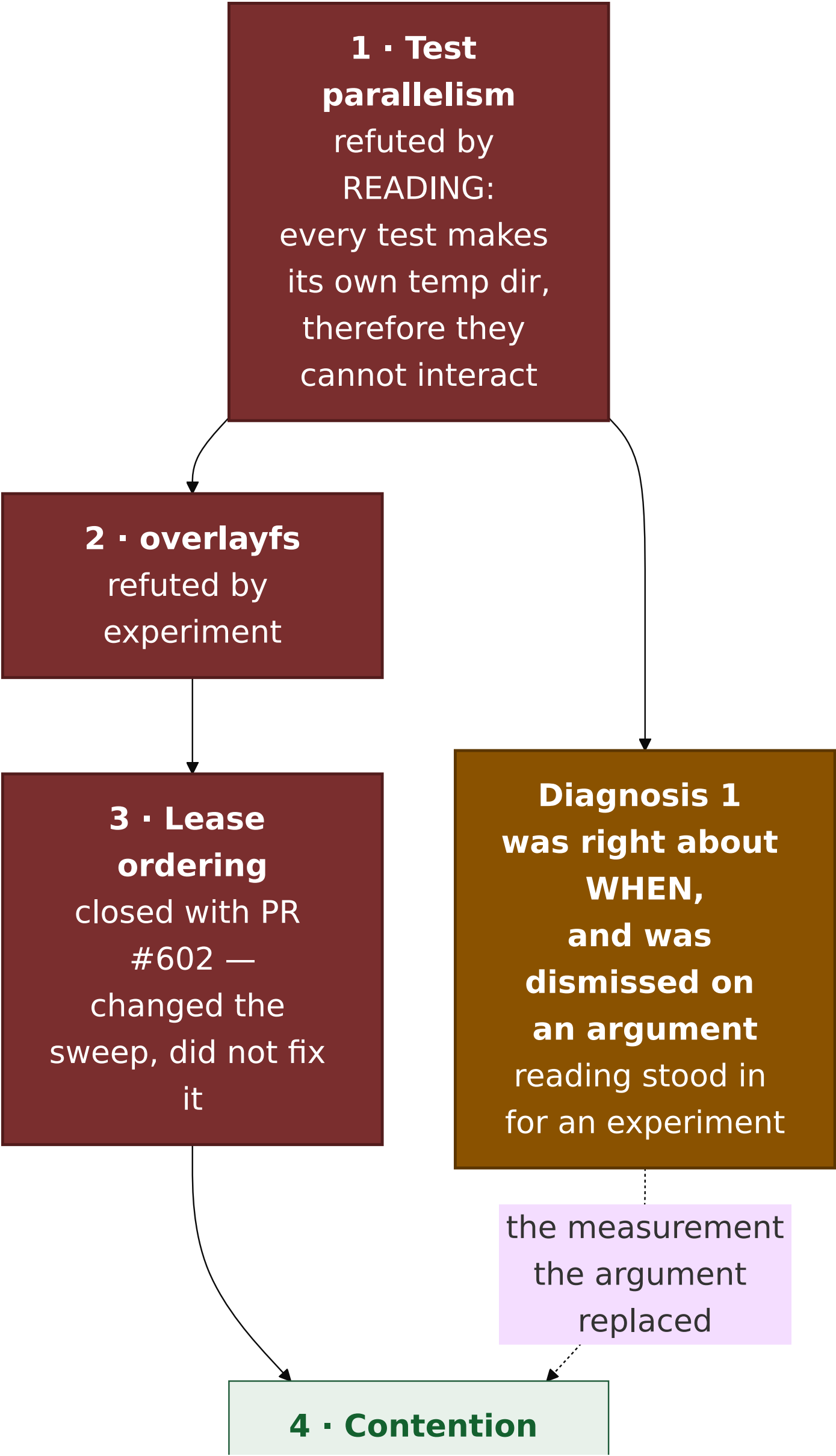
Three tests failed intermittently in CI and blocked two unrelated pull requests, #597 and #616. All three failed on the same kind of assertion — the one each carries as a guard against a sweep that has quietly stopped deleting anything:

the sweep answered but reclaimed nothing — this test would pass against a sweep that had stopped working entirely

The issue was filed as three distinct failures. It was one.

Four diagnoses, three of them wrong

This defect has an unusually bad record, and the record is the reason the decision below is written the way it is.



on the lease
the shape the
measurement
supports

The first diagnosis said "test parallelism" and was refused by reading the code. The reading was correct as far as it went — each test really does make its own temp directory — and the conclusion drawn from it was wrong. Reading was the wrong instrument.

The reproduction

The variable that matters is **running the whole preview:: module in one process**, not the single test:

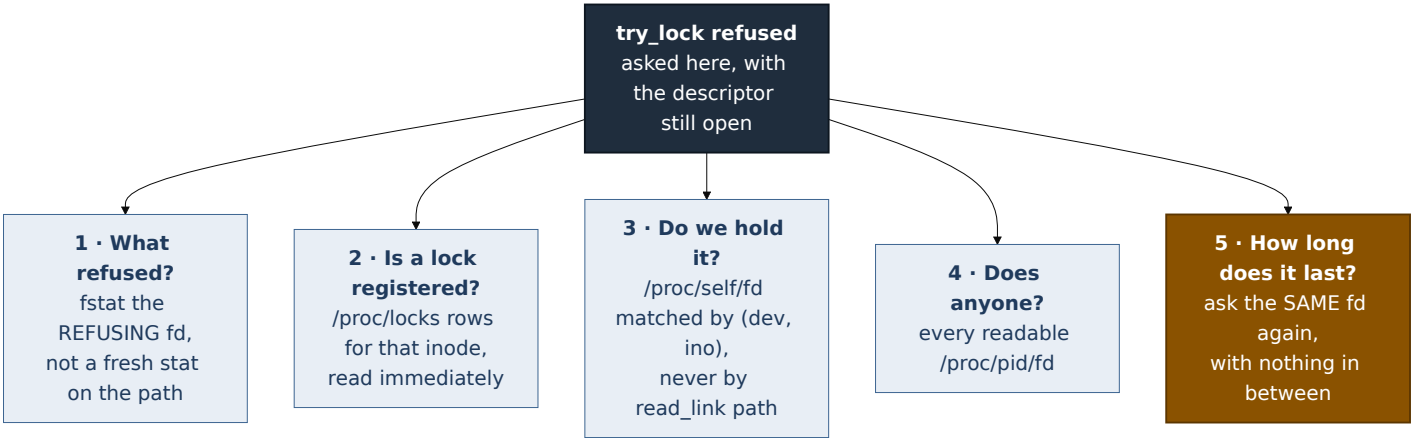
Configuration	Failures
the named test alone, 120 consecutive runs	0 / 120
whole preview:: module, --test-threads=1	0 / 10
whole preview:: module, default 20 threads	12 / 20

So parallelism is necessary and sufficient, and it was measured rather than argued.

The contradiction, and the probe that resolved it

strace put the failure squarely on gate 5: flock(22, LOCK_EX|LOCK_NB) = -1 EAGAIN. Something really was refusing. But every probe run from the test body afterwards showed **no lock and no holder** — an empty /proc/locks, an empty descriptor table.

Those probes ran milliseconds too late. So the probe moved into abandoned_store_lease's error arm, firing at the instant of refusal with the refusing descriptor still open. It asked four questions:



Matching by (dev, ino) rather than by comparing read_link output was deliberate: flock is per-inode, so a holder that reached the same file by a different spelling is invisible to a path comparison.

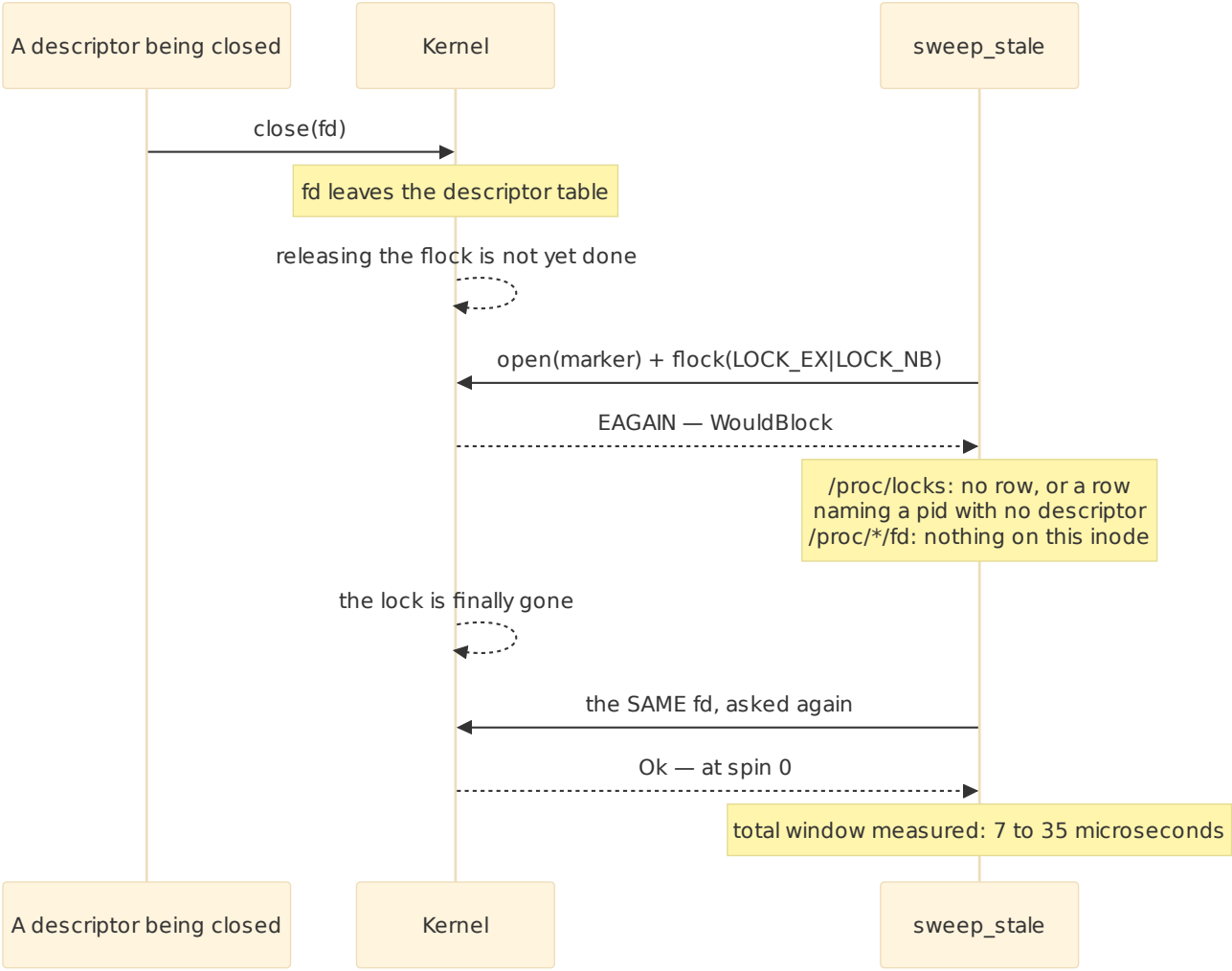
The first version of this probe could not answer question 4. It skipped the cross-process scan whenever it found an in-process descriptor on the inode — and the descriptor try_lock had just refused on is, by construction, always one. Every refusal looked self-explained and the scan never ran once. That is the same defect class this repository calls a green test that proves nothing, wearing a probe's clothes. Excluding the refusing descriptor from its own answer is what made the measurement below possible.

What the probe measured

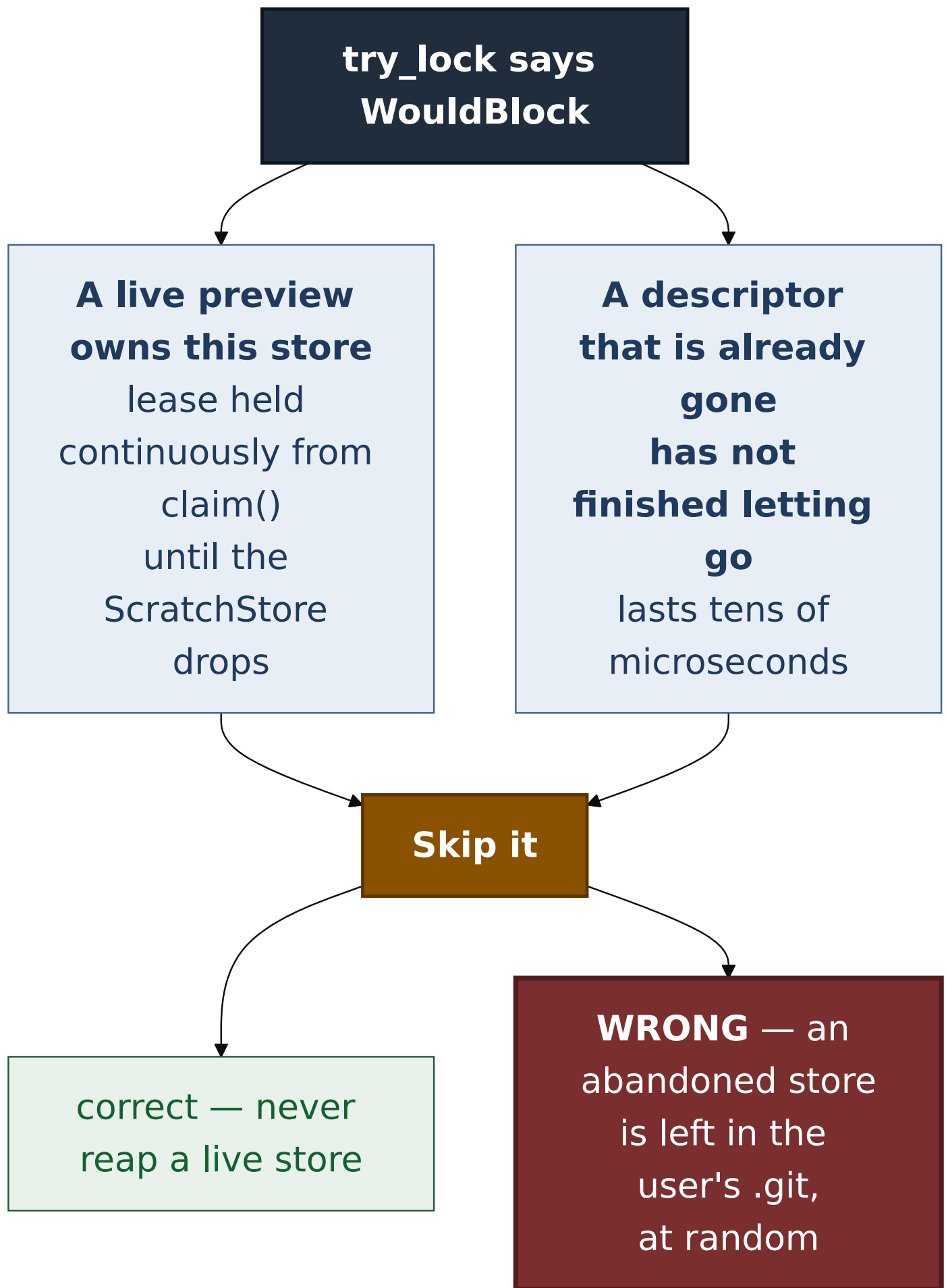
```
LEASE-REFUSAL candidate=.../gv-preview-control thread=None variant=WouldBlock
dev=66310 ino=25562238 locks=[] in_process_fds=[] other_processes=[]
freed_after=27.315µs
LEASE-REFUSAL-SAMEFD ino=25562238 spins_until_free=0
```

- try_lock answered WouldBlock;
- no row in /proc/locks for that inode;
- no descriptor on that inode in this process;
- no descriptor on it in any other process this user can read;
- and the very next try_lock, on the same descriptor with nothing in between, succeeded — at spin 0, in every captured occurrence. The longest window observed was about 35 µs.

Some captures did show a /proc/locks row naming this process while nothing anywhere pointed a descriptor at the inode. A lock with a row and no descriptor is what a close whose lock has not finished being released looks like from outside.



`WouldBlock` therefore conflates two opposite facts:



The sweep was drawing the same conclusion from both, and the harmless-looking half of that conclusion is the leak `sweep_stale` exists to prevent.

Decision

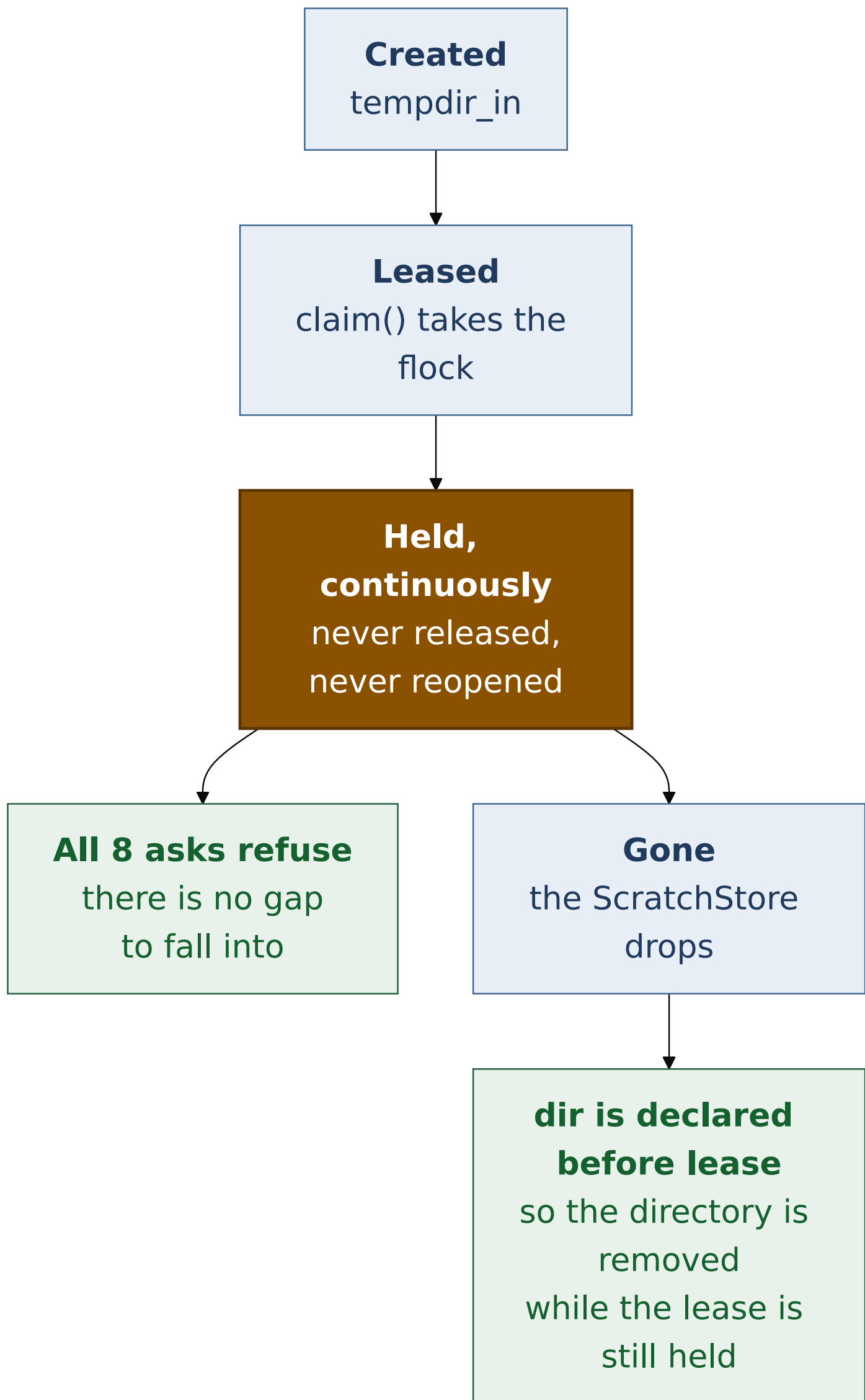
A single non-blocking lock attempt is not evidence that anybody owns a store. The lease gate asks `LEASE_ATTEMPTS` times before it believes a refusal.

```
fn lease_if_free(marker: &std::fs::File) -> bool {
    for attempt in 1..=LEASE_ATTEMPTS {
        if marker.try_lock().is_ok() {
            return true;
        }
        if attempt < LEASE_ATTEMPTS {
            std::thread::sleep(LEASE_RETRY_PAUSE);
        }
    }
    false
}
```

Eight asks, a millisecond apart. Nothing else about the sweep moves: no gate removed, no ordering changed, no new deletion path. Gate 5 answers the same question with more evidence.

Why this cannot reap a live store

This is the load-bearing argument, and it is a property of the design rather than of the timing.



Two orderings carry that guarantee. `claim` takes the lock **before** it writes the magic, so a sweeper arriving mid-creation reads a zero-length file, which is not

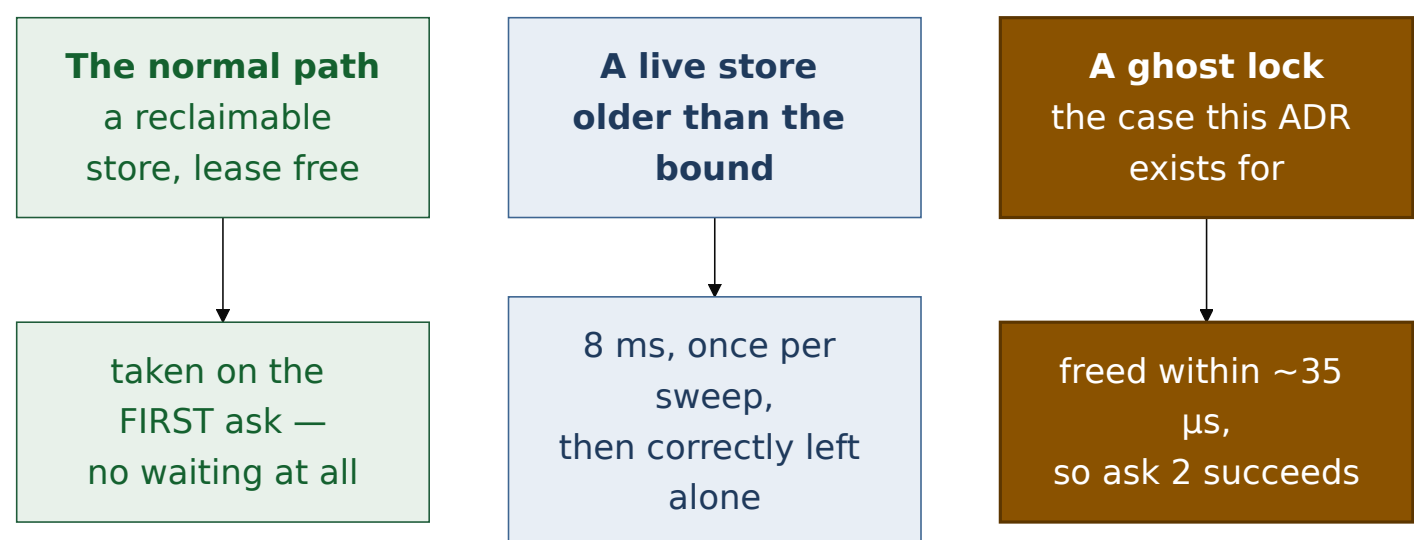
the magic, and leaves at gate 4. And `dir` is declared before `lease` in the struct, so Rust drops the directory while the lease is still held — there is no instant at which a marked store sits on disk with a free lease and a live owner.

A live store's lease is taken by `claim` and held until the `ScratchStore` drops. It is never released and reacquired, so there is no window for a retry to fall into. Every one of the eight asks refuses, and the answer for a live store is byte-for-byte what it was before this change.

Only a lock that is **already gone** can flip from refusing to free — and a lock that is already gone is precisely the definition of an abandoned store. So retrying strictly narrows the set of false negatives and **cannot create a false positive**. That asymmetry is the whole decision.

What it costs, and who pays

The budget is `LEASE_ATTEMPTS × LEASE_RETRY_PAUSE` — at most 8 ms — and it is spent only by a candidate that has already passed gates 1 through 4 and then been refused once.



The free path pays nothing, and that is pinned by its own test rather than left to inspection — a gate that always spent its budget would put 8 ms in front of every reclaimable store in a user's `.git`, and it would still look correct.

Alternatives considered

Fix the tests instead of the sweep. Rejected. The tests assert the sweep's documented contract: an abandoned, marked, unleased store older than the bound must be reclaimed. That guarantee genuinely fails today. Relaxing the tests would have made the symptom go away and left a sweep that skips real leftovers at random — a green suite over a live defect, which is the failure mode this repository has recorded more than any other.

Block on the lock (`LOCK_EX` without `LOCK_NB`). Rejected outright. A blocking acquire is exactly the wedge that `a_named_pipe_wearing_the_markers_name_cannot_wedge_the_sweep` exists to prevent: the sweep runs from `ScratchStore::new` on a spawned task, so anything that can block for ever parks a runtime worker and takes every later preview against that repository with it.

Ignore `WouldBlock` and delete anyway. Rejected, and it is worth naming because it is the mutation the live-store test is built to catch. It removes the only thing separating a running preview from a corpse and hands a live store to `remove_dir_all`.

Drop the lease gate and rely on age alone. Rejected for the same reason gate 5 was added: a two-hour preview is not a corpse.

Distinguish the two `WouldBlock` causes at the syscall level. There is nothing to read. The kernel reports one `EAGAIN`; `/proc/locks` was empty in most captures and, when it was not, named a process holding no descriptor. The retry does not need to tell the causes apart — it needs only to ask again, which the asymmetry above makes safe.

Consequences

- The sweep reclaims abandoned scratch stores deterministically. A leftover in a user's `.git` is no longer skipped on a coin-flip.
- Three intermittently-failing tests are one fixed defect. `#597` and `#616` lose a CI tax they were paying for a bug in neither.
- The sweep can now block its caller for up to 8 ms per stale, live candidate. It already blocked for `read_dir` and `remove_dir_all`; this is a bounded addition to an existing cost, not a new kind of one.
- `LEASE_ATTEMPTS` and `LEASE_RETRY_PAUSE` are sized against measurement — roughly two hundred times the longest observed window — and both are pinned by tests that fail if either is changed.
- **A general caution now has a worked example in this codebase:** an advisory-lock probe that answers "somebody holds this" is answering a question about *this instant*, not about ownership. Anywhere else that pattern appears, one refusal is a hint, not a verdict.

Verification

- **20 / 20 green** on the exact configuration that was 12 / 20 red: the whole `preview::` module in one process, ext4 `TMPDIR`, default 20 test threads.
- `./dev gate` green — full workspace tests, clippy, `cargo fmt --check`, `trunk build`, and 82 browser tests.

Mutation matrix

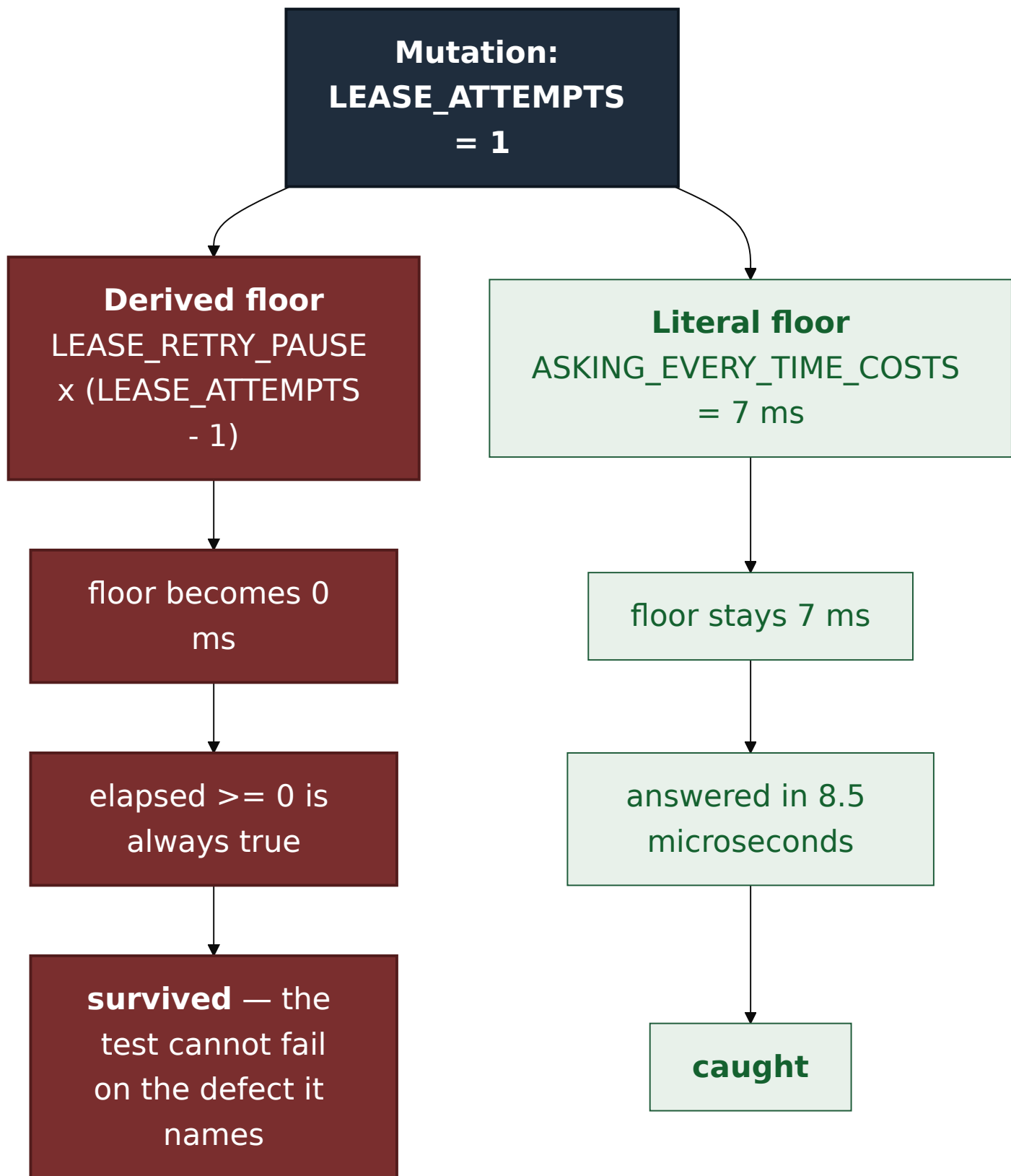
`failure-atlas mutation_check` against committed HEAD `9a9194ff`, working tree reported clean. Every baseline was green and every mutated leg reached and failed an assertion; no compiler failure is counted as a catch.

Test	Mutation	Result	Failed at
a_free_lease_is_taken_on_the_first_ask	remove: the gate never calls <code>try_lock</code> and always returns <code>false</code>	caught (record 191)	the answer
a_free_lease_is_taken_on_the_first_ask	weaken: the successful ask stops returning early, so the gate answers correctly but always spends its whole budget	caught (record 192)	the timing bound — 7.42 ms against 4
a_held_lease_is_refused_only_after_every_ask	remove: <code>LEASE_ATTEMPTS = 1</code> , the shape this code had before #598	caught (record 193)	the timing floor — 8.5 μs against 7 ms
a_held_lease_is_refused_only_after_every_ask	weaken: the refusal arm returns <code>true</code> , so a held lease reads as free	caught (record 194)	the answer

Each pair fails **differently** — one at the boolean the gate returns, one at the time it took to return it. That is what says the pair pins the mechanism rather than noticing the same break twice.

The self-referential floor, and why both bounds are literals

The held-lease floor was first written as `LEASE_RETRY_PAUSE * (LEASE_ATTEMPTS - 1)`, which reads well and is worthless:



A bound derived from the constants under test moves with the mutation. Cut `LEASE_ATTEMPTS` to one and the floor drops to zero, every elapsed time clears it, and record 193 would have read `survived` — the test asserting a mapping by calling the function that defines it, which this repository has shipped before and has a standing rule against.

Both bounds are therefore literals: 7 ms for the held case, 4 ms for the free one. Changing either constant is *meant* to land on these assertions. The failure message from record 193 shows the trap being avoided in the act — "answered in 8.5µs, under the 7ms that asking **1** times costs".